



CS 329S: Machine Learning Systems Design

Stanford, Winter 2022

Schedule & syllabus

The lecture slides, notes, tutorials, and assignments will be posted online here as the course progresses.

Lecture times are **3:15 - 4:45pm PST**. All deadlines are at **11:59pm PST**.

This schedule is subject to change according to the pace of the class.

See *Past course* (<https://stanford-cs329s.github.io/2021/syllabus.html>) for the last year's lectures.

Date	Description	Materials
Mon Jan 3	Understanding machine learning production	Lecture note (https://docs.google.com/document/d/1C3dILmFdYHJmACV/kz99ISTUPF4XQbWb_Ah7mPE12lgo/edit?usp=sharing) Lecture slides (https://docs.google.com/presentation/d/1bhjgRelQ0O5FnYCOGiCVWg_SkRcZ9bffQsgk6yAaL0/edit?usp=sharing) Lessons learned from 150 ML models at Booking.com (https://blog.acolyer.org/2019/10/07/150-successful-machine-learning-models/)
Wed Jan 5	ML and Data Systems Fundamentals	Lecture note (https://docs.google.com/document/d/10K3pYTNvreVy5hl2EqWf_LX3mMW4CQw1TdMrHpIMu00/edit?usp=sharing) Lecture slides (https://docs.google.com/presentation/d/1BYxwxJCb7onDemOtAZTmMc50V3tF80BflkzKUCBLUxg/edit?usp=sharing) Case study: Predict Value of Homes On Airbnb (https://medium.com/airbnb-engineering/using-machine-learning-to-predict-value-of-homes-on-airbnb-9272d3d4739d) Breakout exercise: Designing Twitter's Trending Hashtags (https://mlops-discord.github.io/blog/designing-twiters-trending-hashtags-solution/)
Mon Jan 10	Training Data	Lecture note (https://docs.google.com/document/d/1eiNkA-e3h5NxYMVGMJLTrkrPArC8r1JTV0GKUOgnZfIQ/edit?usp=sharing) Lecture slides (https://docs.google.com/presentation/d/1OYjrmhSBu3Poo5FcY6Wywpu_eR7mtkpe1r8nbbWvArg/edit?usp=sharing)
Wed Jan 12	Feature engineering	Lecture note (https://docs.google.com/document/d/1N7dRx5zwnyXCI0H-VuKpAHTbMBiZXRHzQ7pScSdz1s/edit?usp=sharing) Lecture slides (https://docs.google.com/presentation/d/1Gq3VHW-0ci1gTh97OlckrCBqj3qgkjQNV0SO9t42Eyg/edit?usp=sharing)
Mon Jan 17	No class	Martin Luther King, Jr. Day
Wed Jan 19	Model selection, development, and training	Lecture note (https://docs.google.com/document/d/1UqrR4QP_RMZErc7uLMoLwPtzLys6ikgeUXiyIS1WWPE/edit?usp=sharing) Lecture slides (https://docs.google.com/presentation/d/1X_w55MfBhXGQbZkT_fbW9wdNrOs4sOuydRGPUi_yYCo/edit?usp=sharing)
Mon Jan 24	Offline evaluation	Lecture note (https://docs.google.com/document/d/1_kxf0xRAushBEepKh57mHubWCkVuS37iAqwmJcYfoJE/edit?usp=sharing) Lecture slides (https://docs.google.com/presentation/d/1RqbEbMDmxq53jhjV9V30-DYMv0PIUqINITZsw9Vm9Y/edit?usp=sharing)
Wed Jan 26	Model evaluation Tutorial by Goku Mohandas (MadeWithML) (https://madewithml.com/) RecList (https://arxiv.org/abs/2111.09963) by our very own Chloe He	Goku's ML tutorial (https://docs.google.com/presentation/d/1TIL5MWIVozVtX9384-HSdb7uQVRsQm5amhZ_p39fily/edit?usp=sharing) RecList (https://docs.google.com/presentation/d/e/2PACX-1vQZvsblf-15mwOs-f-UOxY9KoPcUaGS5Uof7i80WYq1MQk5jUDkLjLPWf5fCf3DwxBsEx6QmsLjN0/pub?start=false&loop=false&delayms=3000)
Mon Jan 31	Deployment	Lecture note (https://docs.google.com/document/d/1hNuW6bqWYZjwpiit_8W1cu7Kllb-jTfy3Liof1GJWug/edit?usp=sharing) Lecture slides (https://docs.google.com/presentation/d/1U_zKs19VLJknGE02JDRnrxJ8lgeVF22WSZ_GrA646fY/edit?usp=sharing)
Wed Feb 2	Deployment tutorials How to evaluate MLOps tools by Hamel Husain (https://hamel.dev/) Deploy models with Ray Serve by Simon Mo (https://www.linkedin.com/in/simon-mo-834217162/) (Anyscale)	Hamel's slides (https://docs.google.com/presentation/d/1BVWmp7pfEp6_ffilcvz--_KbFqK-c9bElfbuxHFWPKo/edit?usp=sharing) Hamel's video (https://youtu.be/GHk5HMW4XMA) Ray tutorials (https://github.com/anyscale/academy/blob/main/ray-serve/e2e/tutorial.ipynb)
Mon Feb 7	Diagnosis of ML system failures & data distribution shifts & monitoring	Lecture note (https://docs.google.com/document/d/14uX2m9q7BU_n_mgnM3h6if-s-r0MZrvDb-ZHNjgA1Uyo/edit?usp=sharing) Lecture slides (https://docs.google.com/presentation/d/1tuCibk9Pye-RK1xqiiZXPzT8lIgDUL6CqBkFSYZXkbY/edit?usp=sharing)
Wed Feb 9	Monitoring & Continual Learning Data Distribution Shifts on Streaming Data by Shreya Shankar (https://twitter.com/sh_reya)	Kinbert's slides (https://docs.google.com/presentation/d/1yvbJjukJjXlR6dAVB5dENI_9o5621z7A3P0DF5m4E/edit?usp=sharing) Shreya's slides (slides/cs329s_11_slides_data_distribution_shifts_streams_shreya_shankar.pdf)
Mon Feb 14	Model Deployment @ Stitch Fix by Stefan Krawczyk (https://www.linkedin.com/in/skrawczyk/) Experiment tracking & versioning with Weights & Biases by Lavanya Shukla (https://www.linkedin.com/in/lavanyashukla/)	Stefan's slides (slides/cs329s_12_stefan_model_deployement_stitch_fix.pdf) Lavanya's slides (https://docs.google.com/presentation/d/1aMrSjYUqS5hkT_2Ljs9nhOE12zSS7fe5u9jn_HfcUis/edit?usp=sharing)

Date	Description	Materials
Wed Feb 16	Monitoring Tutorial WhyLogs tutorials by Alessya Visnjic (https://twitter.com/zalessya) Evidently tutorials by Emeli Dral (https://twitter.com/emelidral)	Evidently notebook (https://github.com/evidentlyai/evidently/blob/main/examples/data_stories/bicycle_demand_monitoring_setup.ipynb) WhyLogs notebook (https://colab.research.google.com/drive/1rY9Z_pV8HIB7R18TNqlwJnJmXikUNNN?usp=sharing) WhyLogs' slides on telemetry for ML (https://docs.google.com/presentation/d/13akwLYq8ncEX73-ciTp6enGzK3gG8w57luz682pPJL8/edit?usp=sharing)
Mon Feb 21	No class	Presidents' Day
Wed Feb 23	Deploying time series forecasting and graph neural networks by Kyle Kranen (https://www.linkedin.com/in/kyle-kranen/) ML beyond accuracy: Fairness, Security, Governance by Sara Hooker (https://www.sarahooker.me/)	Sara's slides (https://docs.google.com/presentation/d/1cshMKKSX24L0RL7LNzyOkZnQHD7N-Zyff8iffLlIVYM/edit?usp=sharing)
Mon Feb 28	ML Infrastructure and Platform	Lecture slides (https://docs.google.com/presentation/d/1b-NLyZW2M8D7r_SXLlpLWnosgryMp7luXcn4d9KAn9M/edit?usp=sharing)
Wed Mar 2	Final project discussion	
Mon Mar 7	Integrating ML into business Grace Isford (https://www.linkedin.com/in/graceisford/) (Lux Capital): How to pitch Nnamdi Iregbulem (https://www.linkedin.com/in/nnamdiiregbulem/) (Lightspeed): Go to market Astasia Myers (https://www.linkedin.com/in/astasiamyers/) (Quiet): Business values of AI	
Wed Mar 9	Final project demo day	Recordings of 24 demos can be found on YouTube (https://www.youtube.com/watch?v=AZNTqytOhXk&t=12771s) Join us!